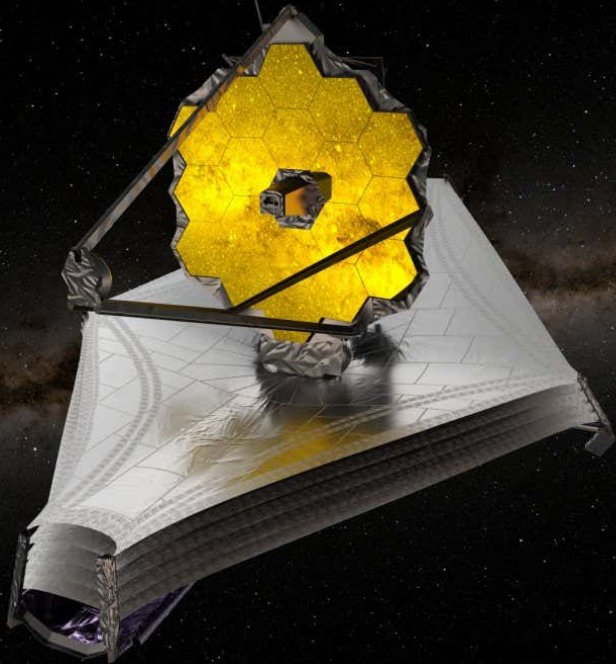




The James Webb Space Telescope

“Hubble 2.0”

Will Handley
wh260@cam.ac.uk

Gonville & Caius College
Cavendish Astrophysics Group

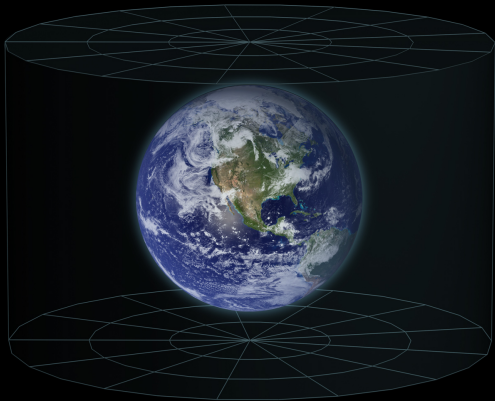
22nd October 2023

Disclaimers

- ▶ I am not on the JWST team
 - ▶ like every astronomer it will continue to have a large impact for many decades
- ▶ JWST has been taking images for half a year now, but only a fraction of them are public
- ▶ This talk is designed to give you a framework to interpret any news stories
- ▶ Hope for a more new-science focussed talk next year.
- ▶ Structure of talk
 1. Orienting our place in the universe
 2. Description of the JWST observatory
 3. Discussion of the science it is interested in
 4. Latest pictures from JWST

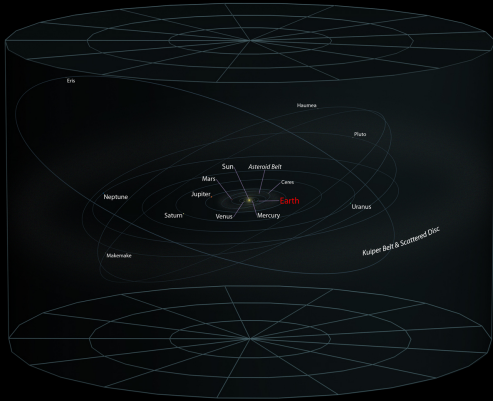
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EARTH

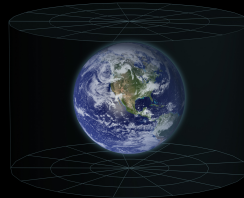


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SOLAR SYSTEM

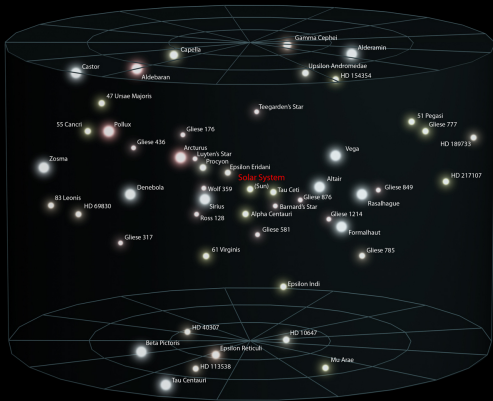


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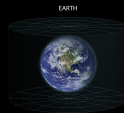


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INTERSTELLAR NEIGHBORHOOD

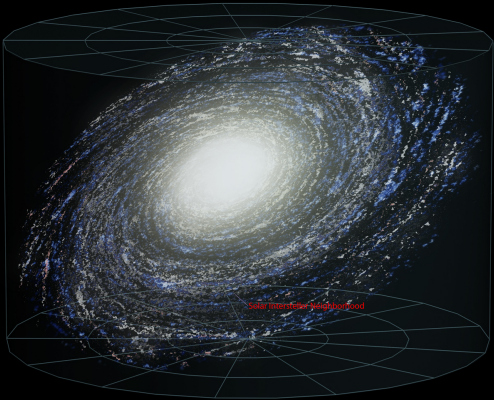


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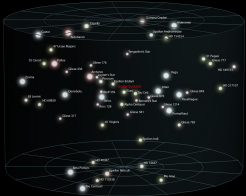


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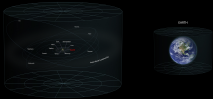
MILKY WAY GALAXY



INTERSTELLAR NEIGHBORHOOD

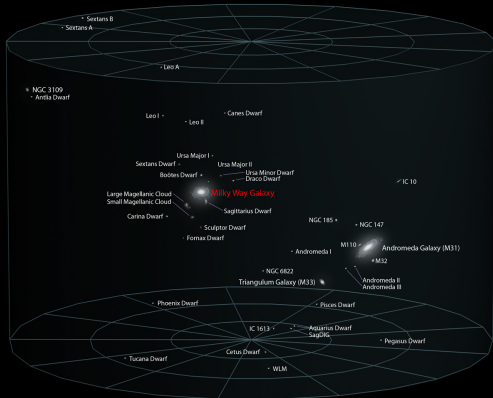


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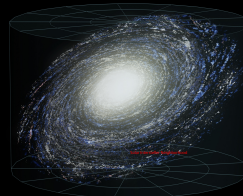


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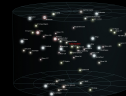
LOCAL GALACTIC GROUP



MILKY WAY GALAXY



INTERSTELLAR NEIGHBORHOOD

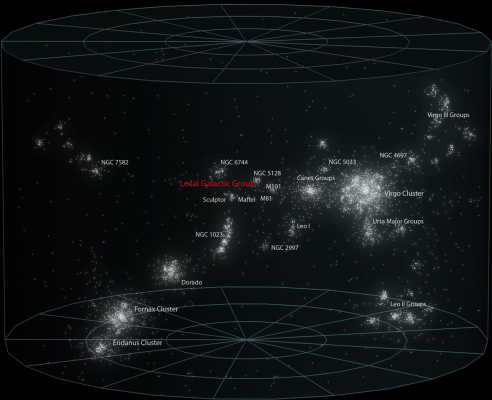


EARTH'S NEIGHBORHOOD



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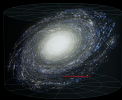
VIRGO SUPERCLUSTER



LOCAL GALACTIC GROUP

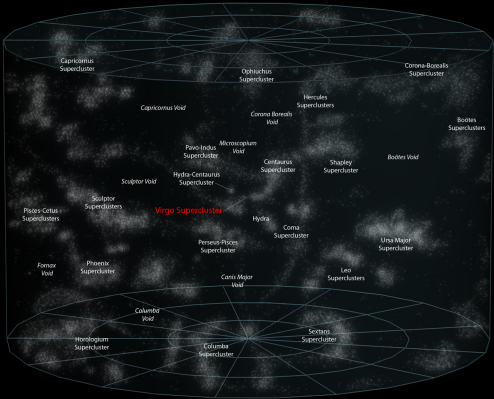


MILKY WAY GALAXY

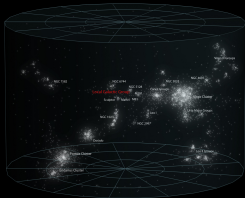


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LOCAL SUPERCLUSTERS



VIRGO SUPERCLUSTER

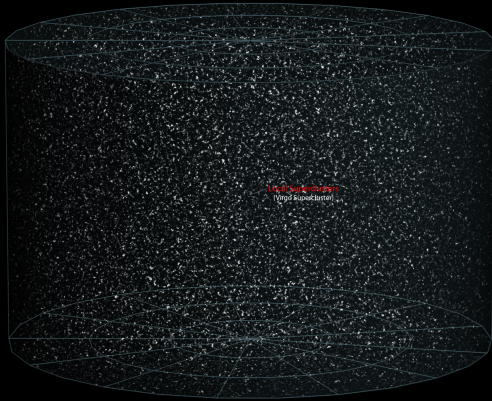


LOCAL GALACTIC GROUP

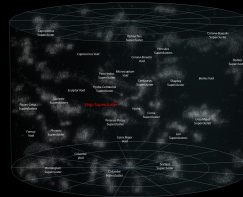


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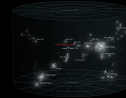
OBSERVABLE UNIVERSE



LOCAL SUPERCLUSTERS

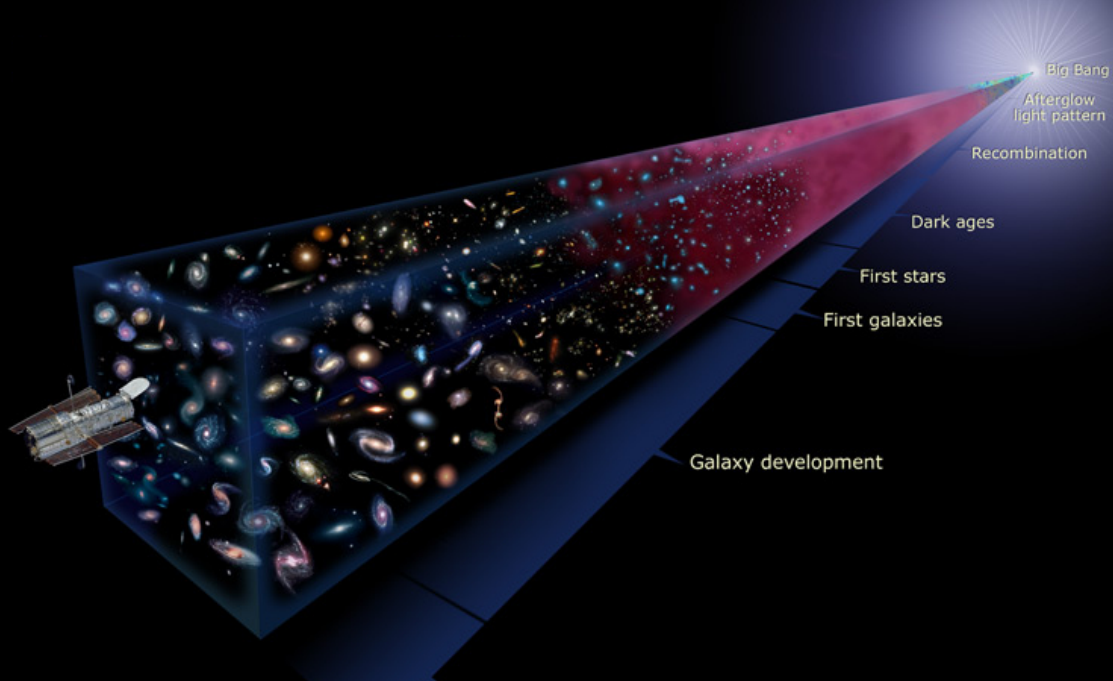


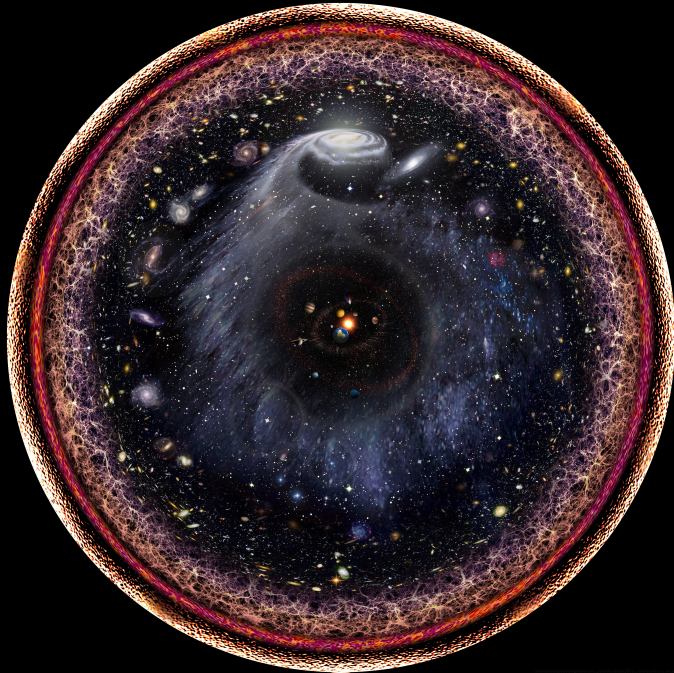
VIRGO SUPERCLUSTER



LOCAL GALACTIC GROUP







Statistics

Where Second lagrange point (L2)

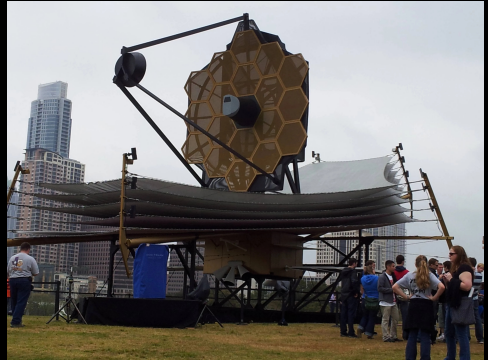
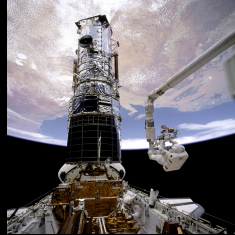
When 5-10 year lifetime

Size ▶ 6.5 meter mirror ($2.5\times$
Hubble [$6\times$ area])
▶ Tennis court sized
sunshield

How Infrared

How much? \$10 billion

How far? ~ 13.6 billion years ago



Webb's Journey from Launch to L2

EARTH

Launch

Launch Vehicle Separates

Distance: ~ 10,000 km
~ 29 min

Solar Array Deploys

Distance: ~ 11,000 km
~ 33 min

Mid-Course Correction

Trajectory Burn
Distance: ~ 133,000 km
~ 14 hours

Sunshield Pallet Deploys

Distance: ~ 454,000 km
~ 3 days

Initial Sunshield Deploys

Distance: ~ 658,000 km
~ 6 days

Sunshield's Layers Separate

Distance: ~ 677,000 km
~ 8 days

Secondary Mirror Unfolds

Distance: ~ 952,000 km
~ 10 days

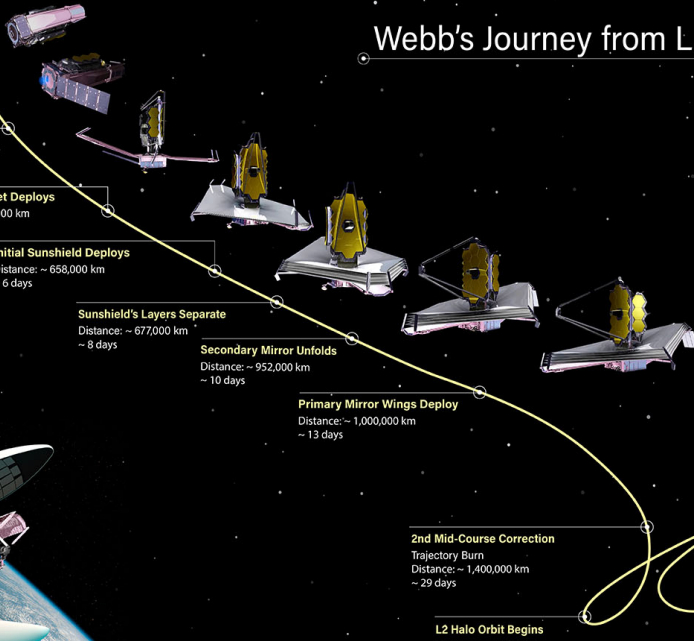
Primary Mirror Wings Deploy

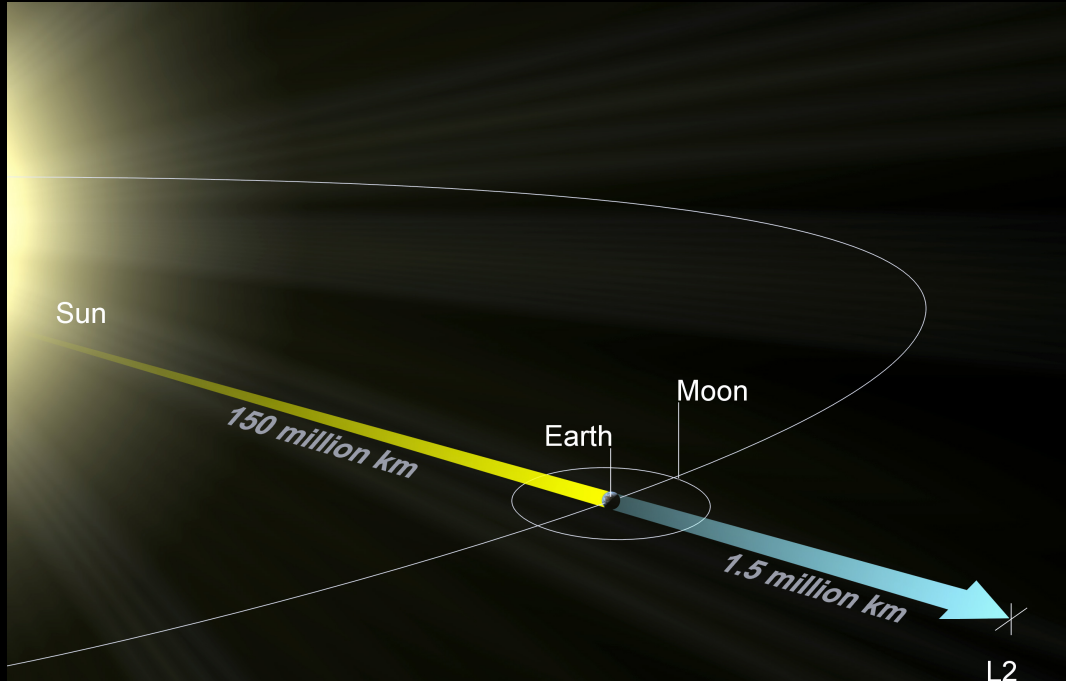
Distance: ~ 1,000,000 km
~ 13 days

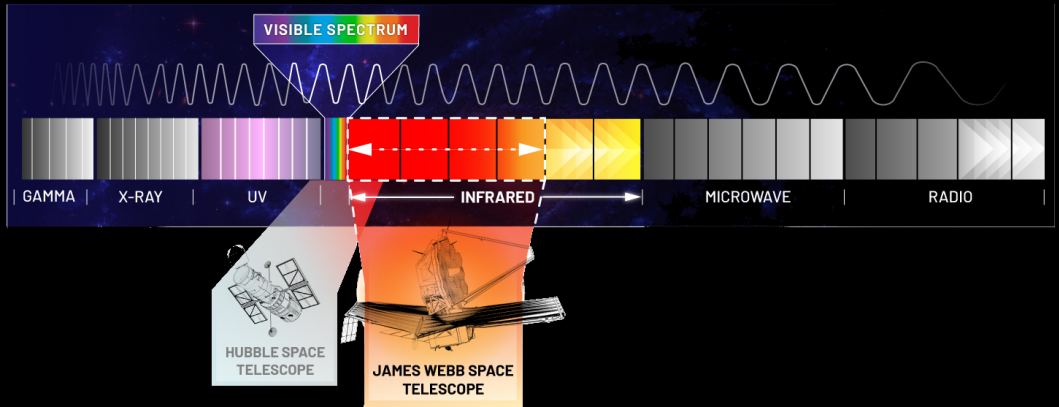
2nd Mid-Course Correction

Trajectory Burn
Distance: ~ 1,400,000 km
~ 29 days

L2 Halo Orbit Begins







Science on the JWST

As a space-based observatory it aims to

- ▶ to search for light from the first stars and galaxies that formed in the universe after the Big Bang
- ▶ to study galaxy formation and evolution
- ▶ to understand star formation and planet formation
- ▶ to study planetary systems and the origins of life

Scientific instruments

- ▶ 17 different “modes” spread across 4 instruments

NIRCam Near Infrared Camera (10×4 MegaPixels HgCdTe)

- ▶ Most of front-page images are with this in its camera mode
- ▶ Also a spectrometer & coronagraph
- ▶ 29 filters & resolutions
- ▶ Cooled to 37K by sunshield

NIRSpec Near Infrared Spectrograph. (2×4 MegaPixels).

- ▶ Three resolution modes in grating wheel assembly + 5 filters

MIRI Mid-Infrared instrument

- ▶ Cooled to < 7 K by cryocooler
- ▶ Builds on legacy of Hubble + coronagraphy

NIRISS Near Infrared Imager and Slitless Spectrograph

- ▶ Single object spectroscopy
- ▶ Aperture masking interferometry
- ▶ Backup to NIRCam